FL Fraud Detection — Benchmarking Results

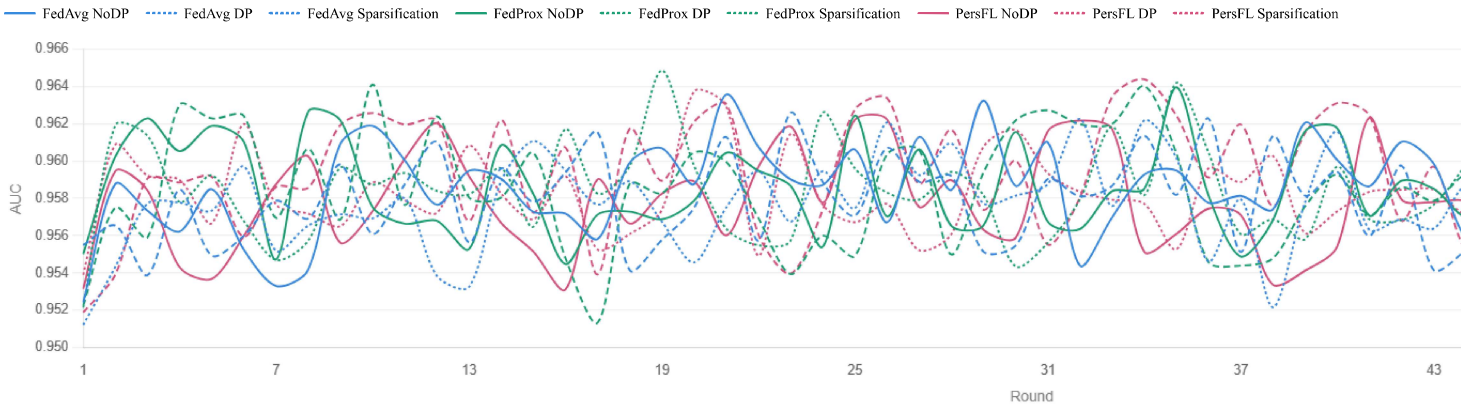
LR model ULB Credit Card Dataset 5 simulated banks 60 rounds 3 algorithms × 3 privacy modes

BEST AUC	BEST F1	AVG JFI (FAIRNESS)	ROUNDS TO ≥95% AUC
0.9648	0.602	0.9999	1
FedProx + Sparsification	PersFL + NoDP	All configs near-perfect	All 9 configurations
Round 19	Round 54		

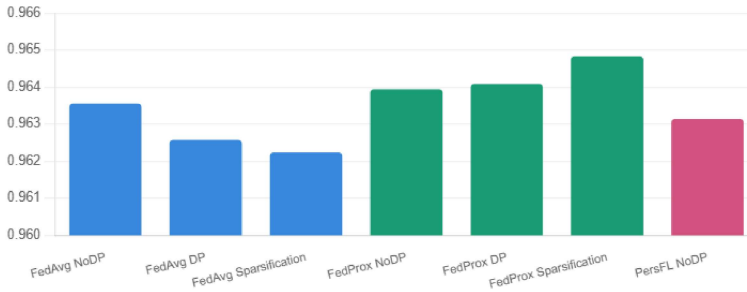
AUC / F1 / LOSS OVER ROUNDS

AUC F1 Loss

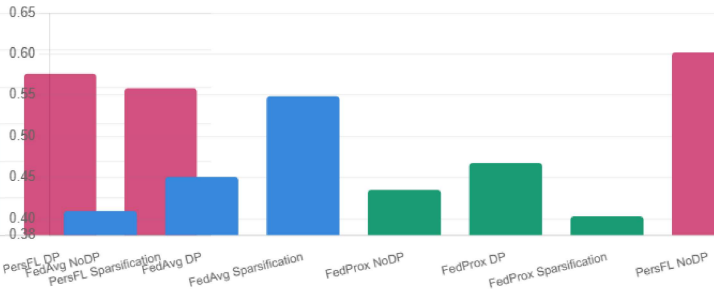
Algorithm: FedAvg FedProx PersFL Privacy: NoDP DP Sparsification



BEST AUC BY CONFIG



BEST F1 BY CONFIG



LEADERBOARD — ALL 9 CONFIGURATIONS

#	ALGORITHM	PRIVACY	BEST AUC	FINAL AUC	BEST F1	Σ(AUC)	JFI	TIME (S)
🥇	FedProx	Sparsification	**0.9648**	0.9575	0.4028	0.00946	0.999908	71.6
🥈	PersFL	DP	**0.9644**	0.9619	0.4591	0.00977	0.999902	70.2
🥉	FedProx	DP	**0.9641**	0.9634	0.4677	0.01028	0.999891	71.9
4	PersFL	Sparsification	**0.9640**	0.9619	0.4339	0.00990	0.999899	71.1
5	FedProx	NoDP	**0.9639**	0.9601	0.4350	0.00983	0.999900	72.3
6	FedAvg	NoDP	**0.9636**	0.9603	0.4094	0.01029	0.999891	72.5
7	PersFL	NoDP	**0.9631**	0.9600	0.6021	0.00992	0.999899	71.3
8	FedAvg	DP	**0.9626**	0.9576	0.4508	0.00965	0.999904	71.9
9	FedAvg	Sparsification	**0.9622**	0.9549	0.5489	0.01099	0.999875	71.8